

Exercise - Week 11: MATH1061/7861

Duration: 30 minutes

3 Questions - 10 marks

Topics: The Definition of a Group, Elementary Group Properties,
Definitions and Examples of Fields, Introduction to counting

Student Name:

Student number: |_|_|_|_|_|_|_|_|_|

*Your student number should be 8 digits and **not** start with an s. e.g. 41234567*

- (1) (4 marks) Determine which of the following is a subgroup of $(\mathbb{Z}_8, +)$. Justify your answer.

Recall we write 0 for the equivalence class $[0]$ and similarly for other elements. In this notation, $\mathbb{Z}_8 = \{0, 1, 2, 3, 4, 5, 6, 7\}$.

(a) $\{0\}$.

(b) $\{0, 3\}$.

(c) $\{0, 2, 4, 6\}$.

- (2) **(3 marks)** Solve the equation $10x + 9 = 6$ in the field $(\mathbb{Z}_{17}, +, \times)$. Show your working clearly.

- (3) **(3 marks)** How many arrangements of the letters in “MATHEMATICS” begin and end with the same letter?